Observe that, defining

$$\boldsymbol{G} = \begin{bmatrix} 0 \\ \boldsymbol{p} \end{bmatrix}$$

yields the kinetic temperature observable $k_{\mathrm{B}}\mathrm{T} = \mathrm{Av}_{\beta}(\boldsymbol{p}^T \boldsymbol{M}^{-1}\boldsymbol{p})/\mathrm{N}_{\mathrm{c}}$, whereas

$$\boldsymbol{G} = \begin{bmatrix} \nabla U(\boldsymbol{q}) \\ 0 \end{bmatrix}$$

gives the *configurational* temperature observable (6.5).

An even simpler configuration-dependent choice is derived from

$$\boldsymbol{G} = \begin{bmatrix} \boldsymbol{q} \\ 0 \end{bmatrix},$$

corresponding to

$$k_{\mathrm{B}}\mathrm{T} = \frac{\mathrm{Av}_{\beta}(\boldsymbol{q} \cdot \nabla U)}{\mathrm{N}_{\mathrm{c}}}.$$

An important point is that the formula of Rugh may not be robust from a computational perspective in cases where the denominator may be close to zero. The simple choice of the kinetic temperature is often desirable since it works for both the infinite domain and for periodic boundary conditions and is such that the denominator is a nonzero constant.

Rugh's principle gives generalized temperature indicators on an infinite domain, but as we have already pointed out, typical molecular dynamics models are confined by periodic boundary conditions to a torus in configuration space (or a toric-cylinder in phase space). Similar results may be obtained in this case. The key point in the proof above is the vanishing contribution from the boundary as $R \to \infty$; in the case of periodic boundary conditions, the boundary terms from opposite sides of the domain can be seen to cancel out, leaving the desired relation. In general one should ensure that the formula for temperature is invariant under translations of the periodic domain along coordinate directions in the configurational domain. If $\boldsymbol{G}$ and H are both periodic in $\boldsymbol{q}$ and $\boldsymbol{G}$ is smooth, then everything works since the outward pointing normal will be in opposite directions on each side of the simulation cell, and the boundary term will cancel.

Thus for periodic boundary conditions we may choose

$$\boldsymbol{G}(z) = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix},$$

or

$$\boldsymbol{G}(z) = \begin{bmatrix} \nabla U(\boldsymbol{q}) \\ \mathbf{0} \end{bmatrix}.$$

However $\boldsymbol{G}(z) = \begin{bmatrix} \boldsymbol{q} \\ \mathbf{0} \end{bmatrix}$ is problematic for periodic boundary conditions.

6.2 Stochastic Differential Equations for Molecular Models

We now have established a framework for the thermodynamic properties of a model system. What would be desirable is that we could approximate the force terms arising in the bulk, without the need to simulate them directly. It should be apparent that constant-energy dynamics (designed to sample the microcanonical ensemble with constant energy E) will not sample the canonical distribution in the absence of the heat bath: such Newtonian trajectories cannot access regions of the phase space where $H(z) \neq H(z_0)$, where z_0 is the initial condition.

On the other hand, changing our perspective to think of trajectories as general tools for sampling a defined distribution, we may consider an alternative dynamics designed specifically to sample the canonical distribution. We refer to a dynamical system that samples ρ_β as a "thermostat," or "thermostatted molecular dynamics" since the canonical ensemble is characterized by a fixed temperature. In principle, even grossly non-physical dynamics would be permissable, if the goal is only to compute stationary (equilibrium) averages. (In Chap. 8, we take up the issue of comparison of dynamics computed using different thermostat methods.) We might also think of a thermostat as a "governor" in the tradition of the governors used in steam engines to regulate the engine temperature. (It is interesting to note that J.W. Gibbs, one of the fathers of modern statistical mechanics, had himself designed a governor for steam engines.) Because it is very difficult (in some sense, probably impossible) to build ergodic thermostats using deterministic dynamical systems, which are, until now, the only tool available to us, we need to develop the theory of stochastic differential equations and, in particular, the application of that theory in the setting of molecular dynamics. This is a fairly tall order and to achieve it in a compact presentation, we assume that the reader is familiar with basic concepts from probability theory (see Appendix B for a quick summary of the main principles).

6.2.1 Random Walks and Wiener Processes

In 1827 Robert Brown noticed that small particles in a fluid were in constant motion, following irregular paths.[5] The essential properties of Brownian motion

[5] Brown was not in fact the only–or even the first–to observe this: Jan Ingenhousz had observed a similar property of coal dust particles on the surface of alcohol already in 1785.